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Valve with configurable default position

Abstract

A valve including a housing, a motor within the housing, and a pinout configuration with an electrical contact. The electrical contact is configured to receive and transmit signals. The valve also includes a motor controller that receives power at the valve, receives a control message, and characterizes the control message as a first type. In response to characterizing the control message as the first type, the motor controller controls the motor to set the valve to a first position in response to determining that the electrical contact is electrically closed and controls the motor to set the valve to a second position in response to determining that the electrical contact is electrically open.

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Plast-O-Matic Valves, Inc. Series EBVA Electric Actuators for 2-Way and 3-Way Ball Valves w/ Direct Manual

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/323,866, filed on Mar. 25, 2022, the entire content of which is hereby incorporated by reference.

FIELD OF THE INVENTION

(1) The present invention relates to a valve, and more specifically, a valve having a configurable default, error position.

SUMMARY

(2) Valves are used to regulate, direct, or control the flow of a fluid and/or gas by opening, closing, or partially obstructing various passageways within a housing of the valve. Vehicles typically include a plurality of valves to control the movement of various fluids throughout the vehicle. Valves are configured to be driven by a motor to direct the flow of fluid and/or gas through the various passageways. In the event of a control failure, valves may be set to a default position to regulate fluid, however, known designs may be met with position customization and power consumption constraints.

(3) The disclosure provides, in one aspect, a valve including a housing, a motor within the housing, and a pullout configuration with an electrical contact. The electrical contact receives and transmits signals. The valve also includes a motor controller that receives power at the valve, receives a control message, and characterizes the control message. In response to characterizing the control message as a first type, the motor controller controls the motor to set the valve to a first position in response to determining that the electrical contact is electrically closed and controls the motor to set the valve to a second position in response to determining that the electrical contact is electrically open.

(4) The disclosure provides, in another aspect, a method for controlling a position of a valve. The method includes receiving power at the valve, receiving a control message, and characterizing, via the motor controller, the control message as a first type. In response to characterizing the control message as the first type, the method includes controlling, via the motor controller, the motor to set the valve to a first position in response to determining that an electrical contact is electrically closed and controlling, via the motor controller, the motor to set the valve to a second position in response to determining that the electrical contact is electrically open.

(5) Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A illustrates a three-way valve including an electrical connector, according to one embodiment.
- (2) FIG. 1B is a cross-sectional view of the three-way valve of FIG. 1A, according to one embodiment.
- (3) FIG. 2A illustrates a four-way valve including an electrical connector, according to one embodiment.
- (4) FIG. 2B is a cross-sectional view of the four-way valve of FIG. 2A, according to one embodiment.
- (5) FIG. 3A illustrates a three-way valve including an electrical connector, according to one embodiment.
- (6) FIG. 3B is a cross-sectional view of the three-way valve of FIG. 3A, according to one embodiment.
- (7) FIG. 4 is a block diagram of a system for controlling a valve, according to one embodiment.

(8) FIG. 5A is a cross-sectional view of the valve of FIG. 4 in a first position, according to one embodiment.
(9) FIG. 5B is a cross-sectional view of the valve of FIG. 4 in a second position, according to one embodiment.

(10) FIG. 6 is a flow chart of a method for controlling an error position of a valve, according to one embodiment.

DETAILED DESCRIPTION OF THE DRAWINGS

(11) Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways.

(12) FIG. 1A illustrates a valve **100**, according to some embodiments. In the illustrated embodiment, the valve **100** is a three-way valve. The valve **100** includes a housing **105**, a main chamber (FIG. 1B) within the housing **105**, a first tube **110**, a second tube **115A**, and a third tube **115B**. The first tube **110**, the second tube **115A**, and the third tube **115B** are in communication with the main chamber of the housing **105** to define passageways for fluid and/or gas to flow through the valve **100**. The valve **100** also includes an electrical port **120**. In the illustrated embodiment, the electrical port **120** includes a pinout configuration **125** with a plurality of electrical connectors (further described below in reference to FIGS. 4-5B). In some embodiments, the valve **100** is mechanically coupled to an external device via the electrical port **120** and is electrically coupled to the external device via the pinout configuration **125**. The pinout configuration **125** receives and transmits control signals from the external device to the components of the valve **100**. In some embodiments, the pinout configuration **125** receives and transmits power to the valve **100**. In some embodiments, the valve **100** is positioned within a fluid system and is fluidly coupled to a pump. The pump may be positioned within a vehicle system. In other embodiments, the valve **100** is physically and/or operatively connected the pump, a fan, a compressor, or any other suitable component of the vehicle system mechanically via the electrical port **120**. The valve **100** is physically and/or operatively connected the pump, a fan, a compressor, or any other suitable component of the vehicle system electrically via the pinout configuration **125**.

(13) FIG. 1B is a cross-sectional view of the valve **100**, according to some embodiments. In the illustrated embodiment, the first tube **110**, the second tube **115A**, and the third tube **115B** are in communication with a main chamber **130** of the valve **100**. The main chamber **130** may be similar to the main chamber described above in reference to FIG. 1A. The main chamber **130** houses interior components of the valve **100**. The valve **100** also includes a central component **135** within the main chamber **130**. The central component **135** is actuated by a motor (further described below in reference to FIG. 4) to rotate the central component **135** and direct the flow of fluid and/or gas from the first tube **110**, to the second tube **115A**, and/or to the third tube **115B**. For example, the central component **135** is positioned to direct the flow of fluid and/or gas from the first tube **110** to the second tube **115A**. In some examples, the central component **135** is positioned to direct the flow of fluid and/or gas from the first tube **110** to the third tube **115B**. In other examples, the central component **135** is positioned to direct the flow of fluid and/or gas from the second tube **115A** and the third tube **115B**. In the illustrated embodiment, the central component **135** is a rotary ball. In other embodiments, the valve **100** could be reconfigured to include elements of other types of valves such as, but not limited to, a poppet valve, a check valve, a thermal mixing valve, etc.

(14) FIG. 2A illustrates a valve **200**, according to some embodiments. In the illustrated embodiment, the valve **200** is a four-way valve. The valve **200** may include similar components to the valve **100** described above with reference to FIG. 1A. As illustrated in FIG. 2A, the valve **200** includes a housing **205**, a main chamber (FIG. 2B) within the housing **205**, a first tube **210**, a second tube **215A**, a third tube **215B**, and a fourth tube **215C**. The first tube **210**, the second tube **215A**, the third tube **215B**, and the fourth tube **215C** are in communication with the main chamber of the housing **205** to define passageways for fluid and/or gas to flow through the valve **200**. The valve **200** also includes an electrical port **220**. In the illustrated embodiment, the electrical port **220** includes a pinout configuration **225** with a plurality of electrical connectors (further described below in reference to FIGS. 4-5B). In some embodiments, the valve **200** is mechanically and electrically coupled to an external device via the electrical port **220** and the pinout configuration **225**. The pinout configuration **225** receives and transmits control signals from the external device to the components of the valve **200**. In some embodiments, the pinout configuration **225** receives and transmits power to the valve **200**. In some embodiments, the valve **200** is positioned within fluid system and is fluidly coupled to a pump. The pump may be positioned within a vehicle system. In other embodiments,

the valve **200** is physically and/or operatively connected the pump, a fan, a compressor, or any other suitable component of the vehicle system mechanically via the electrical port **220**. The valve **200** is physically and/or operatively connected the pump, a fan, a compressor, or any other suitable component of the vehicle system electrically via the pinout configuration **225**.

(15) FIG. **2B** is a cross-sectional view of the valve **200**, according to some embodiments. In the illustrated embodiment, the first tube **210**, the second tube **215A**, the third tube **115B**, and the fourth tube **115C** are in communication with a main chamber **230** of the valve **200**. The main chamber **230** houses interior components of the valve **200**. The valve **200** also includes a central component **235** within the main chamber **230**. The central component **235** is actuated by a motor (further described below in reference to FIG. **4**) to rotate the central component **235** and direct the flow of fluid and/or gas from the first tube **210**, to the second tube **215A**, the third tube **215B**, and/or the fourth tube **215C**. For example, the central component **235** is positioned to direct the flow of fluid and/or gas from the first tube **210** to the second tube **215A**. In some examples, the central component **235** is positioned to direct the flow of fluid and/or gas from the first tube **210** to the third tube **215B**. In some examples, the central component **235** is positioned to direct the flow of fluid and/or gas from the first tube **210** to the fourth tube **215C**. In other examples, the central component **235** is positioned to direct the flow of fluid and/or gas between any of the second tube **115A**, the third tube **115B**, and the fourth tube **115C**. In the illustrated embodiment, the central component **235** is a rotary ball. In other embodiments, the valve **200** could be reconfigured to include elements of other types of valves such as, but not limited to, a poppet valve, a check valve, a thermal mixing valve, etc.

(16) FIG. **3A** illustrates a valve **300**, according to some embodiments. In the illustrated embodiment, the valve **300** is a vertical three-way valve. It should be understood that the valve **200** may include similar components to the valve **100** described above with reference to FIG. **1A**. As illustrated in FIG. **3A**, the valve **300** includes a housing **305**, a main chamber (FIG. **3B**) within the housing **305**, a first tube **310**, a second tube **315A**, and a third tube **315B**. The first tube **310**, the second tube **315A**, and the third tube **315B** are in communication with the main chamber of the housing **305** to define passageways for fluid and/or gas to flow through the valve **300**. The valve **300** also includes an electrical port **320**. In the illustrated embodiment, the electrical port **320** includes a pinout configuration **325** with a plurality of electrical connectors (further described below in reference to FIGS. **4-5B**). In some embodiments, the valve **300** is mechanically and electrically coupled to an external device via the electrical port **320** and the pinout configuration **325**. The pullout configuration **325** receives and transmits control signals from the external device to the components of the valve **300**. In some embodiments, the pullout configuration **325** receives and transmits power to the valve **300**. In some embodiments, the valve **300** is positioned within fluid system and is fluidly coupled to a pump. The pump may be positioned within a vehicle system. In other embodiments, the valve **300** is physically and/or operatively connected the pump, a fan, a compressor, or any other suitable component of the vehicle system mechanically via the electrical port **320**. The valve **300** is physically and/or operatively connected the pump, a fan, a compressor, or any other suitable component of the vehicle system electrically via the pinout configuration **325**.

(17) FIG. **3B** is a cross-sectional view of the valve **300**, according to some embodiments. In the illustrated embodiment, the first tube **310**, the second tube **315A**, and the third tube **315B** are in communication with a main chamber **330** of the valve **300**. The main chamber **330** houses interior components of the valve **300**. The valve **300** also includes a central component **335** within the main chamber **330**. The central component **335** is actuated by a motor (further described below in reference to FIG. **4**) to rotate the central component **335** and direct the flow of fluid and/or gas from the first tube **310**, to the second tube **315A**, and/or the third tube **315B**. For example, the central component **335** is positioned to direct the flow of fluid and/or gas from the first tube **310** to the second tube **315A**. In some examples, the central component **335** is positioned to direct the flow of fluid and/or gas from the first tube **310** to the third tube **315B**. In other examples, the central component **335** is positioned to direct the flow of fluid and/or gas between the second tube **315A** and the third tube **315B**. In the illustrated embodiment, the central component **335** is a rotary ball. In other embodiments, the valve **300** could be reconfigured to include elements of other types of valves such as, but not limited to, a poppet valve, a check valve, a thermal mixing valve, etc.

(18) FIG. **4** is a block diagram illustrating a system for controlling a valve **400A**. The valve **400A** may be any one of the valves **100**, **200**, **300**, as described above. Although illustrated alone, the valve **400A** is positioned within fluid system and is fluidly coupled to a pump. The pump is positioned within a vehicle system such as, for example, a cooling system of the vehicle. The vehicle system may encompass various types and designs of vehicles. For example, the vehicle may be an automobile, an electric automobile, a

motorcycle, a truck, a bus, and others. The valve **400A** includes a motor controller **430** and a motor **435** in electrical contact with the pullout configuration (e.g., the pinout configuration **125**, **225**, **325**). The motor **435** is configured to rotate the central component of the valve **400A** between at least a first position and a second position. The motor **435** is an electrical motor, such as but not limited to a direct-current motor operable at variable speeds. In some embodiments, the motor **435** is a brushless direct-current (BLDC) motor. In other embodiments, the motor **435** can be a variety of other types of motors, including but not limited to a brush DC motor, a stepper motor, a synchronous motor, or other direct-current or alternating-current motors.

(19) The motor controller **430** is electrically connected to the motor **435** and provides one or more control signals to operate the motor **435**. The motor controller **430** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the motor controller **430** and/or the motor **435**. For example, the motor controller **430** includes, among other things, a processing unit (e.g., a microprocessor, a microcontroller, or another suitable programmable device) and a memory unit. In some embodiments, the motor controller **430** is implemented partially or entirely on a semiconductor (e.g., a field-programmable gate array ["FPGA"] semiconductor, an application specific integrated circuit ["ASIC"], or other programmable semiconductor devices as appropriate for a given application) chip, such as a chip developed through a register transfer level ("RTL") design process. In one example, upon receiving a control signal, the motor controller **430** controls and/or operates the switching of a plurality of electronic switches (e.g., FETs), in order to selectively drive the motor **435** at a speed and/or direction. In some embodiments, the motor controller **430** and the motor **435** form a single unit. In other embodiments, the motor controller **430** and the motor **435** are individual components of the valve **400A**.

(20) The valve **400A** includes a plurality of electrical connectors (e.g., inputs, outputs, input/outputs [e.g., general purpose input/output "GPIO"] etc.). In some embodiments, the plurality of electrical connectors (for example, electrical contacts) includes a battery positive connector A, a battery negative (e.g., ground) connector B, a control signal connector C, an enable signal connector D, and a failsafe (e.g., an error operation) connector E.

(21) FIGS. **4**, **5A**, and **5B** illustrate the pullout configuration which includes a plurality of electrical connectors A-E (e.g., pins, inputs, outputs, input/outputs, etc.). The single pin connection of the failsafe connector E is advantageous for allowing more options for electrical connection selection, a configurable default position, and saving space. In some embodiments, the plurality of electrical connectors A-E includes the battery positive connector A, the battery negative (e.g., ground) connector B, the control signal connector C, the enable signal connector D, and the failsafe connector E. Although shown in FIGS. **4-5B** as connectors A, B, C, D, and E, each connector of the pinout configuration may be assigned to any one of the associated connectors shown in any order. The control signal connector C is configured to be electrically connected to a system controller **450** via electrical contact with a system controller bus **455**. In reference to FIG. **4**, although the control signal connector is shown as connector C, the control signal connector C may be any one of the plurality of connectors. The system controller bus **455** is configured to be in electrical contact with the valve **400A**, a first device **400B**, and a second device **400C**. The first device **400B** and the second device **400C** are other components of the vehicle system, for example, a compressor, a fan, or any suitable vehicle component. In some embodiments, the first device **400B** and the second device **400C** include a similar pinout configuration to the valve **400A**. The system controller bus **455** may use one or more known control bus protocols, such as a Controller Area Network (CAN) bus protocol, a Local Interconnect Network (LIN) bus protocol, or other control bus protocol types as required for a given application. In some embodiments, the system controller **450** may be configured to communicate using multiple control bus protocols.

(22) The system controller **450** can include a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the system controller **450** and/or the valve **400A**. For example, the system controller **450** includes, among other things, a processing unit (e.g., a microprocessor, a microcontroller, or another suitable programmable device) and a memory. In some embodiments, the system controller **450** is implemented partially or entirely on a semiconductor (e.g., a field-programmable gate array ["FPGA"] semiconductor, an application specific integrated circuit ["ASIC"], or other programmable semiconductor devices as appropriate for a given application) chip, such as a chip developed through a register transfer level ("RTL") design process.

(23) The memory includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as read-only memory ("ROM"), random access memory ("RAM") (e.g., dynamic RAM ["DRAM"], synchronous DRAM ["SDRAM"], etc.), electrically erasable programmable read-only memory ("EEPROM"), flash memory, a

hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The illustrated processing unit is connected to the memory and executes software instructions that are capable of being stored in a RAM of the memory (e.g., during execution), a ROM of the memory (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. Software included in some implementations of the valve **400A** can be stored in the memory of the system controller **450**. The software can include, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The system controller **450** is configured to retrieve from memory and execute, among other things, instructions related to the control processes and methods described herein. In other constructions, the system controller **450** includes additional, fewer, or different components.

(24) In operation, the system controller **450** outputs a control signal to the motor controller **430**. The motor controller **430** receives the control signal and operates the motor **435** based on the control signal. In some embodiments, the control signal is a pulse-width modulated signal. The pulse-width modulated signal can have a duty cycle (e.g., 10%, 50%, 100%, etc.). In some embodiments, the duty cycle corresponds to an operating speed of the motor **435** (e.g., 10% of full speed, 50% of full speed, 100% of full speed, etc.).

(25) With continued reference to FIGS. 4-5B, the control signal connector C connects to the system controller bus **455** of the vehicle for communication of control protocol of the valve **400A**. In one embodiment, the system controller **450** and system controller bus **455** are the same as a CAN controller and CAN bus and/or LIN controller and LIN bus. In other embodiments, the system controller **450** and system controller bus **455** are separate from the vehicle's CAN controller and CAN bus and/or LIN controller and LIN bus. In some embodiments, the control signal connector C may connect directly to a CAN bus or LIN bus of the vehicle. In some embodiments, the LIN bus may be a sub-bus of the CAN bus of the vehicle. The failsafe connector E may be connected to a ground connection (e.g., the battery negative connector B or other ground connection) or be left open (e.g., not electrically connected to an external connection). The connection of the failsafe connector E determines the failsafe position (e.g., error position) of the valve **400A** in the event of control failure (i.e., no control signal from the system controller **450** is received via the control signal connector C or a loss of power occurs). If the failsafe connector E is connected to a ground connection, the failsafe position is set to a first position (FIG. 5A). If the failsafe connector E is left open, the failsafe position is set to a second position (FIG. 5B). Therefore, the failsafe position can be changed based on the electrical connection of the failsafe connector E. The configurable failsafe position allows for very low power consumption when the valve **400A** is not in motion as no power is drawn to maintain the valve **400A** in the failsafe position (i.e., no power is consumed when the valve **400A** is not changing positions). In the event of control failure, the valve **400A** returns to the configured default position (e.g., the failsafe position). The battery positive connector A is electrically connected to a positive terminal of a vehicle battery to provide power to the valve **400A**. In some embodiments, the battery positive connector A is electrically connected to the system controller **450** or the system controller bus **455** to receive power. The battery negative terminal B is electrically connected to a negative terminal of the vehicle battery to ground the valve **400A**. In some embodiments, the battery negative terminal B is electrically connected to a ground terminal of the system controller **450** or the system controller bus **455** to ground the valve **400A**. The enable signal connector D is electrically connected to the system controller **450** or the system controller bus **455** to allow the control signal connector C to receive control signals from the system controller **450**. For example, the enable signal connector D allows control signals to be received and transmitted by the control signal connector C when the valve **400A** is receiving power.

(26) The system controller **450** may further include a communications module. The communications module provides analog and/or digital communications from the system controller **450** to outside devices. In some embodiments, the communications module outputs diagnostic information concerning the system controller **450** and/or other components of the cooling system. The communications module may include an output driver in the form of a digital driver such as SAE J1939, CAN bus, or LIN bus for communicating directly to the vehicle's data bus, or the communications module may generate another suitable analog or digital signal depending on the needs of the specific application. The communication module may further be configured to receive data from one or more data buses within the vehicle, such as SAE J1939, CAN bus, and/or LIN bus.

(27) FIG. 5A is a cross-sectional view of a valve **500** in the first position, according to some embodiments. The valve **500** may be any one of the valves **100**, **200**, **300**, **400A**, as described above. In some embodiments, the first position is based on the electrical connection of the failsafe connector E. For example, the motor controller **430** determines that the failsafe connector E is connected to a ground connection by measuring the

voltage or current at the failsafe connector E. When the motor controller **430** measures zero volts and zero amps at the failsafe connector E, the motor controller **430** determines that the failsafe connector E is connected to a ground connection. When the motor controller **430** determines that the failsafe connector E is connected to a ground connection, the motor controller **430** sets the failsafe position to the first position. In some embodiments, the first position is set by the motor **435** rotating the central component **135**, **235**, **335** to direct the flow of fluid and/or gas from the first tube **110**, **210**, **310** to the second tube **115A**, **215A**, **315A**. FIG. 5B is a cross-sectional view of the valve **500** in the second position, according to some embodiments. In some embodiments, the second position is based on the electrical connection of the failsafe connector E. For example, the motor controller **430** determines that the failsafe connector E is not connected to an external connection by measuring the voltage or current at the failsafe connector E. When the motor controller **430** measures a non-zero voltage and zero amps at the failsafe connector E, the motor controller **430** determines that the failsafe connector E is not connected to an external connection (e.g., open circuit). When the motor controller **430** determines that the failsafe connector E is left open, the motor controller **430** sets the failsafe position to the second position. In some embodiments, the second position is set by the motor **435** rotating the central component **135**, **235**, **335** to direct the flow of fluid and/or gas from the first tube **110**, **210**, **310** to the third tube **115B**, **215B**, **315B**. In some embodiments, the failsafe position may be set to the first position or the second position in the event of control failure.

(28) FIG. 6 is a flow chart of a method **600** for controlling an error position (e.g., failsafe position) of a valve, for example the valve **100**, according to some embodiments. Although the method **600** is described herein with reference to the valve **100**, the method **600** may be executed to control the error position of the valves **200**, **300**, **400A**. The order of the steps disclosed in the method **600** could vary. For example, additional steps may be added to the process and not all of the steps may be required, or steps shown in one order may occur in a second order. In one embodiment, the motor controller **430** is configured to execute the method **600**. In other embodiments, the system controller **450** is configured to execute the method **600** in combination with the motor controller **430**. The method **600** begins at step **605** the valve **100** receives power. For example, the valve **100** receives power via the pinout configuration **125**. The method **600** then proceeds to step **610**.

(29) At step **610**, the valve **100** receives a control message from the system controller **450**. For example, the motor controller **430** receives a control signal indicative of an operation of an operation of the valve **100** from the system controller **450**. The motor controller **430** also characterizes the control message as a first type or a second type. For example, if the control message is the first type (e.g., an invalid control message), the motor controller **430** determines that it has received a control message indicative of power failure, an erroneous control message, or the motor controller **430** does not receive a control message. When the motor controller **430** determines that the control message is the second type (e.g., a valid control message), the method **600** proceeds to step **615**.

(30) At step **615**, the motor controller **430** transmits a control signal to the motor **435** to set the operating position of the valve **100** based on the valid control message from the system controller **450**. For example, the motor controller **430** transmits the control signal to the motor **435** to rotate the central component **135** to the first position. In some embodiments, the motor **435** rotates the central components **135** to the second position based on the control signal. The second position is different from the first position, as described above. In some embodiments, such as embodiments including the valve **200**, the motor **435** rotates the central component **135** to a third position based on the control signal. The third position may direct the flow of liquid and/or gas from the first tube **210** to the fourth tube **215C**. The third positions is different from the first position and the second position, as described above. After the completion of step **615**, the method **600** may return to step **610** to receive a subsequent control message.

(31) When an invalid control message is received at step **610**, the method **600** proceeds to step **620**. At step **620**, the motor controller **430** determines whether the error operation (e.g., failsafe) connector E is electrically open or electrically closed. When the motor controller **430** determines that the error operation connector E is closed (i.e., connected to the ground connection), the method **600** proceeds to step **625**. For example, motor controller **430** measures the voltage and current at the error operation connector E and determines that the error operation connector E is electrically connected to the ground connection. At step **625**, the motor controller **430** controls the motor **435** to set the position of the valve to the first position. For example, the motor controller **430** transmits a control signal to the motor **435** to rotate the central component **135** to the first position. After the completion of step **625**, the method **600** may return to step **610** to receive a subsequent control message.

(32) When the motor controller **430** determines that the error operation connector E is open (e.g., not electrically connected to an external connection) at step **620**, the method **600** proceeds to step **630**. For example, motor controller **430** measures the voltage and current at the error operation connector E and determines that the error operation connector E is not electrically connected to an external connection (e.g., left open). At step **630**, the motor controller **430** controls the motor **435** to set the position of the valve to the second position. For example, the motor controller **430** transmits a control signal to the motor **435** to rotate the central component **135** to the second position. As describe above, the second position is different than the first position. After the completion of step **615**, the method **600** may return to step **610** to receive a subsequent control message. The steps of method **600** may be completed as many times as necessary to continuously set the position of the valve **100**. Accordingly, the method **600** allows for the valve **100** to be set in either position, first or second, in the event of control failure as an error operation. If a valid control message is received by the motor controller **430**, then the motor controller **430** will follow the instruction of the control message.

(33) Although the invention has been described with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described.

Claims

1. A valve comprising: a housing; a motor within the housing; a pinout configuration with an electrical contact, wherein the electrical contact is configured to receive and transmit signals; and a motor controller configured to: receive power at the valve; receive a control message; characterize the control message; and in response to characterizing the control message as a first type: control the motor to set the valve to a first position in response to determining that the electrical contact is electrically closed, and control the motor to set the valve to a second position in response to determining that the electrical contact is electrically open.
2. The valve of claim 1, wherein the second position is different than the first position.
3. The valve of claim 2, wherein the valve includes a central component and when controlling the motor to set the valve to the second position, the motor controller is configured to: control the motor to rotate the central component to the second position.
4. The valve of claim 3, wherein when determining that the electrical contact is electrically open, the motor controller is configured to: determine that the electrical contact is not electrically connected to an external connection.
5. The valve of claim 2, wherein the motor controller is configured to: control the motor to set an operating position of the valve in response to characterizing the control message as a second type, wherein the operating position is based on the control message being the second type, and wherein the operating position is at least one selected from the group consisting of the first position and a second position, the second position different from the first position.
6. The valve of claim 2, wherein the motor controller is configured to: control the motor to set an operating position of the valve in response to characterizing the control message as a second type, wherein the operating position is based on the control message being the second type, and wherein the operating position is at least one selected from the group consisting of the first position, a second position, and a third position, the second position different from the first position and the third position, and the third position different from the first position.
7. The valve of claim 1, wherein the valve includes a central component and when controlling the motor to set the valve to the first position, the motor controller is configured to: control the motor to rotate the central component to the first position.
8. The valve of claim 7, wherein when determining that the electrical contact is electrically closed, the motor controller is configured to: determine that the electrical contact is electrically connected to a ground connection.
9. A fluid system including the valve of claim 1 and including a pump.
10. A vehicle comprising a cooling system including the fluid system of claim 9.
11. A method for controlling a position of the valve of claim 1, the method comprising: receiving power at the valve; receiving the control message; characterizing, via the motor controller, the control message as the first type; and in response to characterizing the control message as the first type: controlling, via the motor

controller, the motor to set the valve to the first position in response to determining that the electrical contact is electrically closed, and controlling, via the motor controller, the motor to set the valve to the second position in response to determining that the electrical contact is electrically open.

12. The method of claim 11, wherein the second position is different than the first position.

13. The method of claim 12, wherein the valve includes a central component and controlling the motor to set the valve to the second position includes: controlling the motor to rotate the central component to the second position.

14. The method of claim 13, wherein determining that the electrical contact is electrically open includes: determining, via the motor controller, that the electrical contact is not electrically connected to an external connection.

15. The method of claim 11, wherein valve includes a central component and controlling the motor to set the valve to the first position includes: controlling the motor to rotate the central component to the first position.

16. The method of claim 15, wherein determining that the electrical contact is electrically closed includes: determining, via the motor controller, that the electrical contact is electrically connected to a ground connection.

17. The method of claim 11, comprising: controlling, via the motor controller, the motor to set an operating position of the valve in response to characterizing the control message as a second type, wherein the operating position is based on the control message being the second type, and wherein the operating position is at least one selected from the group consisting of the first position and the second position, the second position different from the first position.

18. The method of claim 11, comprising: controlling, via the motor controller, the motor to set an operating position of the valve in response to characterizing the control message as a second type, wherein the operating position is based on the control message being the second type, and wherein the operating position is at least one selected from the group consisting of the first position, the second position, and a third position, the second position different from the first position and the third position, and the third position different from the first position.

19. The valve of claim 1, further comprising an electrical port positioned on the housing and configured to couple to an external device, the electrical port including the pinout configuration with the electrical contact.

20. A valve comprising: a housing; a motor within the housing; a pinout configuration with an electrical pin, wherein the electrical pin is configured to receive and transmit signals; and a motor controller configured to: receive power at the valve; receive a control message; characterize the control message; and in response to characterizing the control message as a first type: control the motor to set the valve to a first position in response to determining that the electrical pin is electrically connected to an external connection, and control the motor to set the valve to a second position in response to determining that the electrical pin is not electrically connected to an external connection.
